

Harish Narasimhan K

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LinkedIn | Github | CodeChef | LeetCode

Education

SASTRA University, B.Tech in Computer Science and Engineering
CGPA: 9.25/10.0 (As of 7th Semester)

Oct 2022 - May 2026

Experience

Software Development Engineer Intern, Amazon Web Services (AWS) Jan 2026 – Jun 2026

- Improved **OpenSearch cluster stability** by resolving a critical memory retention issue, reducing memory overhead by over **99%** and preventing cluster manager failures during long-running snapshot operations
- Developed a **two-tier metadata and statistics caching** system for OpenSearch's DataFusion analytics engine, reducing query latency by over **60%**
- Built an end-to-end telemetry pipeline** exposing cache efficiency, query execution, and storage I/O metrics through REST APIs and CloudWatch, enabling real-time monitoring and rapid diagnosis of production issues
- Engineered an AWS CDK-based OpenSearch benchmarking framework** integrating automated cluster deployment, CloudWatch monitoring, and continuous profiling, reducing environment setup from **hours to minutes** and accelerating performance testing and regression analysis

Backend AI Intern, Insightschakra Technologies Private Limited

May 2025 – Jul 2025

- Developed **AI-powered business intelligence workflows** on Vertex AI, transforming marketing and retail data into **actionable insights** through natural-language exploration of enterprise analytics
- Designed a **multi-agent decision-support system** using Google ADK, combining enterprise data retrieval and LLM reasoning for complex BigQuery analytical workloads
- Optimized agent workflows through **parallel execution**, improving response reliability and reducing end-to-end query latency by **90%**

Projects and Research Works

Anywhere Voyager - AI-Trip-Planner

Github

- Built an **AI-powered travel planning application** using LangGraph and FastAPI, integrating multiple external APIs and agent-based workflows to generate personalized itineraries and travel recommendations

Conversational PDF Query System

Github

- Built a **Retrieval-Augmented Generation (RAG)** system for conversational PDF querying, integrating semantic search, LLM-based reasoning, and voice interactions to reduce query-response time by over **70%**

Block-Based Separable Reversible Data Hiding in Encrypted Images using Homomorphic Encryption

Paper

- Designed and implemented a reversible data-hiding framework using **additive homomorphic encryption**, achieving **75%** embedding capacity improvement and lossless image recovery while preserving data confidentiality

Skills

Languages: Python, Java, Rust, SQL

Distributed Systems & Performance: Concurrency, Caching, JVM Internals, Memory Management, Performance Engineering, Profiling

Cloud & Data Platforms: OpenSearch, AWS CDK, EC2, S3, CloudWatch, Vertex AI, BigQuery, Google ADK

Generative AI & ML: Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Natural Language Processing, Supervised Learning, Deep Neural Networks

Frameworks & Libraries: Git, LangChain, HuggingFace, PyTorch, Scikit-learn, Streamlit

Achievements

- Ranked 12th among 70 teams in the Tredence Infinity AI Hackathon for building a RAG-powered legal assistant using Databricks and Streamlit, enabling case-law retrieval, legal search, and document summarization
- Achieved Knight rating on LeetCode and a 3-star rating on CodeChef